

Math performance prediction & EduCare

(Mental health & well-being Question 4)

Group 26 Introduction – CPES Saclay & Telecom SudParis & M2 Data Science



**6 students with
complementary skills
in Data Science, ML,
Health, Sociology &
Business.**



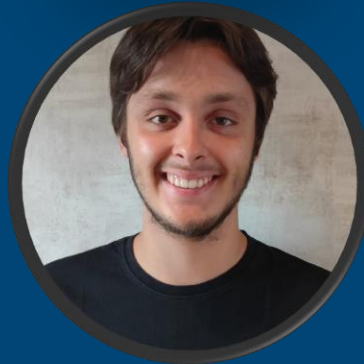
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Data Preparation & Exploration

❑ **Dataset:** PISA Cycles (2015, 2018, 2022) - Unified Dataset (1.7M student records, +250 variables):

Cleaned student-questionnaire data and cognitive test information

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Cleaned student-questionnaire data and cognitive test information

❑ **Exploration findings:**

Missingness is systematic; 1 categorical variable to encode after cleaning : CYC; Correlation between MathScore and ST255 (how many books are in your home) or reading_q6_average score

❑ **Challenges:**

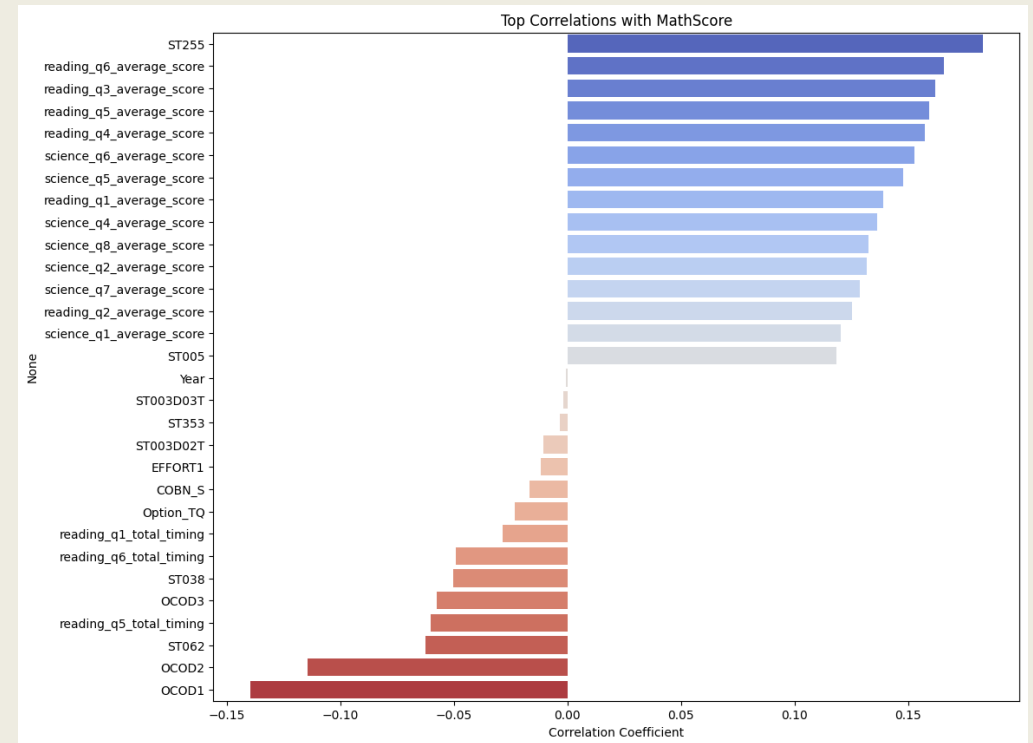
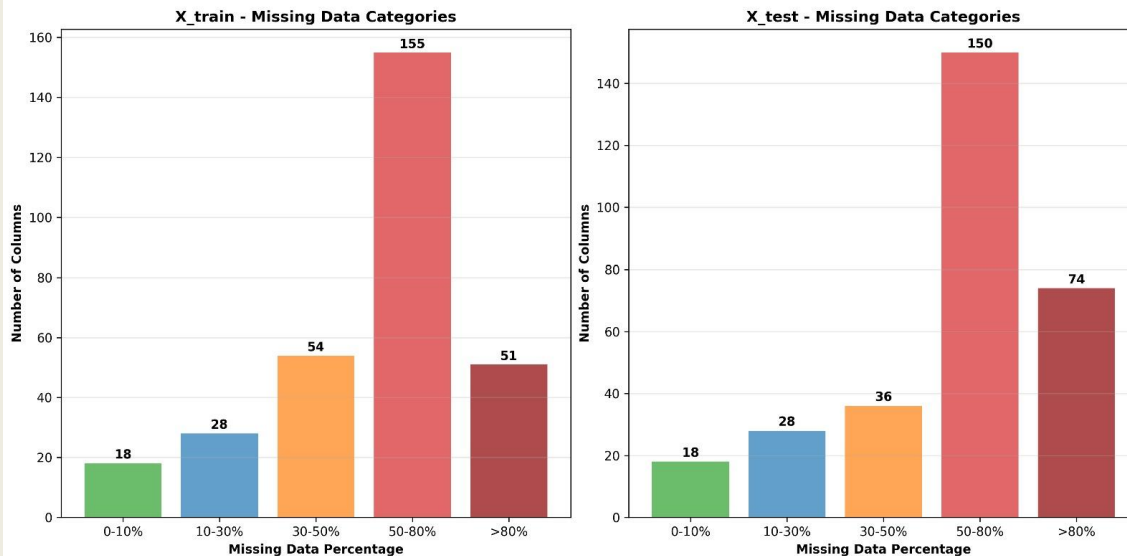
Massive dataset highly fragmented

→ Long training time

~60% overall missing values

} Filtering
+
Imputation

Distribution of Missing Data by Category

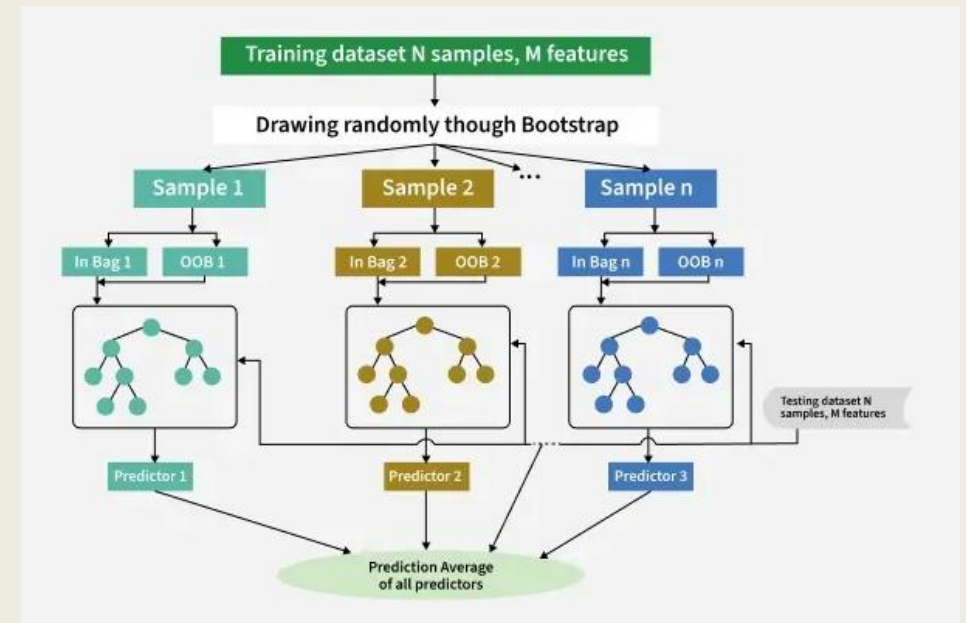


Modeling Approach

- ❑ **Models tested:** Stacking (XGBoost, LightGBM, HistGradientBoostingRegressor), Random Forest, CatBoost, Regressions
- ❑ **Feature selection:** Highly correlated features were removed, reducing the feature set from +250 to 76 key features, to prevent overfitting.
- ❑ **Time management:** GridSearchCV parameters were optimized on a 20% subsample to save time before final model training on the full dataset.
- ❑ **Best model:** CatBoost → better performance, handles categorical and mixed data natively or with less sensitivity
- ❑ **Evaluation metric:** $R^2 \sim 78\%$ on evaluation
- ❑ **Hyperparameter tuning:** learning rate, max depth, n_estimators

Model	R^2	RMSE
Lasso	0.42	95.37
Random Forest	0.92 (1st model overfitted) ~0.43 after being fixed	94.55
Ridge	0.44	93.71
Stacking (submitted)	0.51	85.25
CatBoost (submitted)	0.78	58.73

CatBoost



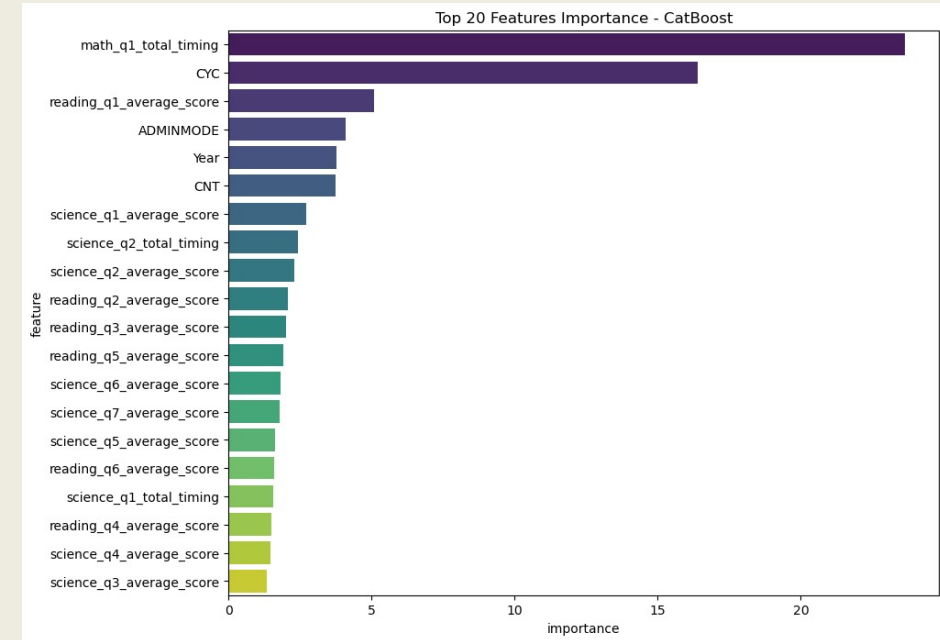
Explainability & Key Drivers

❑ Importance (top features):

Math_q1_total_timing (23.66) / CYC (16.39) / reading_q1_average_score (5.10)

❑ Key insight:

The model relies most on the time taken to answer question 1 in the math test and PISA Assessment Cycle (CYC), indicating that CYC and how long students spend on math tasks strongly influence MathScore. Reading and science average scores also matter, but with smaller impact.

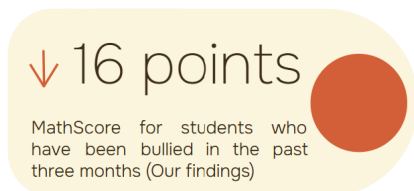


Conclusions & Next Steps

Strenghts	Limitations	Recommendations
- Robust preprocessing pipeline. - Comprehensive model testing. - Strategic optimization for large dataset. - CatBoost achieved high R^2 with low RMSE. - Identification of key explanatory variables.	- Limited model performance (scores of other models restricted by missing values/imputation strategy). - CatBoost = model black box (limited interpretability beyond feature importance data).	- Handle zero-inflated targets: we could use a two-stage model (classify > 0, then regress) to better capture MathScore and reduce RMSE. - Improve sparse feature imputation: maybe we should reconsider dropped features and apply k-NN or MICE instead of simple median/mode fills. - Tune deeper models: we could also try using Bayesian Optimization or Optuna to efficiently search deeper CatBoost/XGBoost configurations for potential R^2 gains.



Mental health & well-being: how can we come up with innovative solutions to support students?



58%

Of students reported being reluctant to seek help for mental health concerns at school (National Institute of Health, USA)

Mental health issues are very prevalent among students all around the world and a few dare to speak to a responsible adult for various reasons. In addition, these issues lead to lower performances during exams.

The education system is unequipped to face these challenges

The current systems put into place to deal with harassment and low self-esteem issues, such as the Phare system in France, are greatly insufficient, often addressing the issue too late, when the student's suffering has become unbearable to them. **EduCare**, by automatically reporting the students showing signs of hardships, allows the student support teams to act rapidly, directly after the first signs of struggles by students.

Additionally, **EduCare** provides a tool for victims and witnesses to report bullying and harassment events directly on their phones, including anonymously, thus lowering the barrier to seeking help.



Our vision

We envision two parts for our application, one for the students to report the issues they are facing and one for the school to monitor the students. This tool will integrate into the current traditional tracking systems at schools, like Moodle or Pronote, by giving a **different perspective** to school administrators and allowing the support needed to be delivered quickly. The app is a way for the education support team to assess and **react to mental health issues in real time** : one of its main goals is to **create a feeling of connection between the students and their supervisors**.

Our product will be marketed for the public or private education organisation as a subscription.

Pictures of the prototype (student support team's view)

